Introduction to Classes & Objects Project

Objectives: ***At the end of this project the student should be able to:***

1. Declare a class and use it to create an object.
2. Declare methods in a class to implement the class's behaviors.
3. Declare instance variables (attributes) in a class.
4. Call an object's method to make that method perform its task.
5. Learn the differences between instance variables of a class and local variables of a method.
6. Use a constructor and initialize objects with constructors.
7. Distinguish between primitive types and reference types.

a) Create a new project, call it "bank", and then add to it the following class:

```
public class Account {
	private String name; //instance variable

	public String getName() {
		return name;
	}
	public void setName(String name) {
		this.name = name;
	}
}
```

b) Create a new class called "AccountTest" that contains a main method. In the main method, create an object of class Account, and print the value of variable "name" of this object.

c) Do we need to use import statement to import "Account" class?
Yes [ ] No [ ] Why?_____

d) Is it necessary to use word *this* in method *setName*?

e) What are the default values for instance variables of the following types

Int	
double & float	
boolean	
Object	

f) Add a constructor that specifies custom initialization for the instance variable "name", now try to run AccountTest again

g) Which of the following statements is correct and which is false:
- If a class does not define constructors, the compiler provides a default constructor with no parameters.
- There is no default constructor in a class that declares constructor.
- The class can have only one constructor.

h) Add another instance variable called "balance", and add setter/getter methods(use Eclipse facility to add set/get methods). Set method should ensure that balance won't be negative.

i) Add another constructor that specify custom initialization for variables: name, balance. The constructor should ensure that balance won't be negative.

j) Add a method called *deposit* that increase the balance, the method should ensure that deposit *amount* is greater than zero.

k) Add a method called *withdraw* that withdraws money from an account. Ensure that withdrawal amount does not exceed the account's balance. If it does, the balance should be left unchanged and the method should print a message indicating " withdrawal amount exceeded account balance".

l) Test these methods in the class AccountTest.

**** Make the code Interactive using Dialog boxes.**